Coupling the Point Model to the Growth Field

Progress since the meeting of 20 October 2026

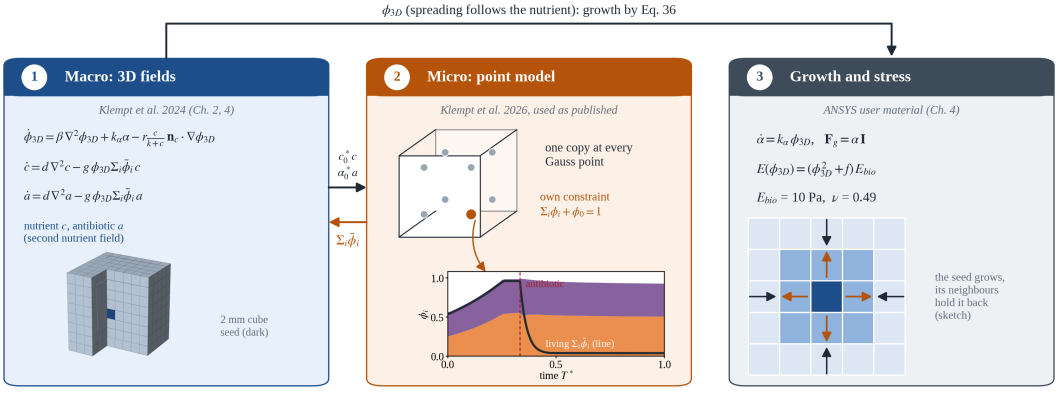
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Where the coupling stands

- Implemented as agreed in your element: the point model (Klempt et al. 2026) runs at every Gauss point as published, consumption by the living biomass $\sum_i \bar{\phi}_i$, antibiotic in the second nutrient field. A Python version solves the same equations.
- Composition is robust: $\sum_i \bar{\phi}_i$, the shares and c agree between the two codes and change by less than 1 % from 16^3 to 128^3.
- The front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$ of Eq. 34 needs care: at the paper's r the cell Péclet number is about **31** on 16^3, so a central gradient oscillates.
- With a second-order scheme the seed von Mises stress converges to 6.2×10^{-4} **Pa** (separate example, from 32^3 on).

The coupling agreed on 20 October



Output: ϕ_i, ψ_i at each Gauss point. $\bar{\phi}_i = \phi_i \psi_i$: only living biomass consumes. Assumed, not from a paper: c^*, α^* linear in time, $\Sigma_i \bar{\phi}_i$ one step late, new points take the neighbours' living composition, ϕ_{min} = 0.001 (converged). s from the data window.

Figure 1: Agreed coupling: the 3D fields of Klempt et al. (2024) pass c^* and α^* to the point model of Klempt et al. (2026) at each Gauss point, which returns $\sum_i \bar{\phi}_i$ for the consumption in Eq. 35.

Separate example with antibiotic

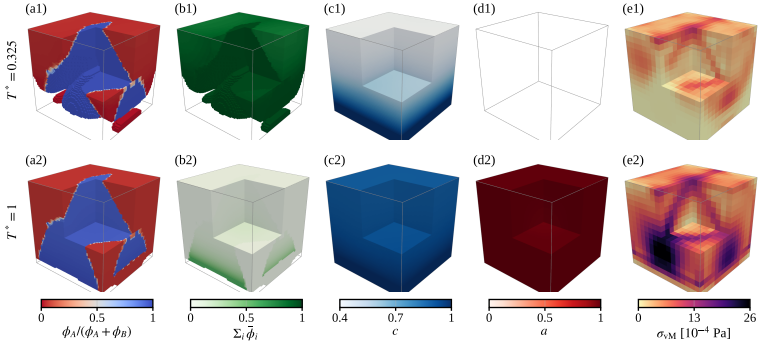
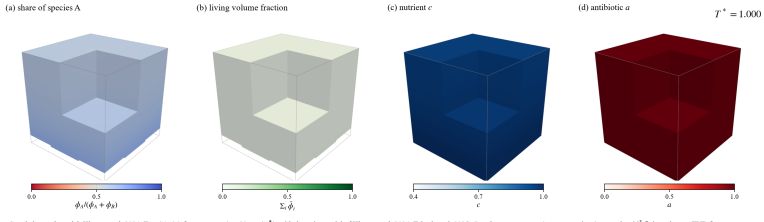


Figure 2: Separate start, $T^* = 0.325$ (top) and $T^* = 1$ (bottom): (a) $\phi_1/(\phi_1+\phi_2)$, (b) $\sum_i \bar{\phi}_i$, (c) c, (d) a, (e) von Mises stress. Python, 64^3, one octant removed.

- A (point-model species 1) in the seed, B (species 2) in four blocks at the z edges; nutrient from the bottom, antibiotic from all outer faces from $T^* = 1/3$ ($\alpha_0^* = 100$)
- Newly reached points take the living composition of their neighbours; the antibiotic kills B and, where the nutrient is low, A

Mixed example with antibiotic



Coupled growth model: Klempt et al. 2024, Eqs. 34–36, front term on ($v = 10$ mm/T^*), with the point model of Klempt et al. 2026 (Fritsch et al. 2025, Case I, true parameters) at every point; 2 mm cube, 64^3 finite volumes, TVD front term.
Mixed start: species A and B in equal amounts together in the centre seed. New biofilm takes the composition of its neighbours.
(a) Share of species A (point-model species 1; B is species 2; blue: A, red: B) and (b) living volume fraction $\Sigma_i \bar{\phi}_i$, $\bar{\phi}_i = \phi_i \omega_i$, both shown where the biofilm $\bar{\phi}_\Sigma > 0.5$.
(c) Nutrient c, held at 1 in the bottom layer $y < -0.75$ mm and consumed by the living biomass. (d) Antibiotic a, held at 1 on the outer shell from $T^* = 1/3$ ($a_0^* = 100$). One octant ($x, y, z > 0$) is removed to show the inside.
Constants: $\beta = 0.02$ mm^2/T^*, $r = 10$ mm/T^*, $K = 1$, $k_a = 10^{-3}/T^*$, $d = 10^3$ mm^2/T^* for c and a, $g\,l^2/d = 1$ for c and a, $\phi_{\min} = 10^{-3}$, $\alpha_0^* = 100$ from $T^* = 1/3$.
Sources: β, r, k_a, d, g from Klempt et al. 2024, Table 2, converted to the 2 mm cube; α_0^* and the onset from Klempt et al. 2026, case 4. $T^* = 1$ is the duration of the point model's data (Fritsch et al. 2025).

Figure 3: Mixed start, $T^* = 1$: (a) $\phi_1/(\phi_1 + \phi_2)$, (b) $\sum_i \bar{\phi}_i$, (c) c, (d) a. Python, 64^3.

- A and B start together in the seed with equal amounts; same nutrient and antibiotic as before
- The antibiotic acts where $c^* = c_0^* c$ falls below $\alpha^* b_i$: the nutrient field decides where it kills

Five examples on the finest grid

Table 1: Seed means at $T^* = 1$ (Python, 64^3, TVD front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$, outer step 0.025).

Example	$\sum_i \bar{\phi}_i$	$\phi_1/(\phi_1+\phi_2)$	c	$\alpha - 1$	von Mises [Pa]
Separate	0.965	0.978	0.536	9.06×10^{-4}	6.56×10^{-4}
Separate, antibiotic	0.152	0.951	0.911	8.95×10^{-4}	5.68×10^{-4}
Mixed, equal amounts	0.961	0.859	0.608	8.33×10^{-4}	3.50×10^{-4}
Mixed, equal amounts, antibiotic	0.038	0.610	0.973	8.30×10^{-4}	3.23×10^{-4}
Mixed (0.25, 0.30), antibiotic	0.037	0.510	0.973	8.30×10^{-4}	3.22×10^{-4}
Grid series $32^3/64^3/128^3$ at outer step 0.0025 with antibiotic: 128^3 value, extrapolated ($\alpha - 1$ and stress in 10^{-4}), GCI					
Separate	0.157	0.952		9.11; 9.22 (1.5 %)	6.28; 6.70 (8.3 %)
Mixed, equal amounts	0.038	0.606		8.32; 8.54 (3.3 %)	3.30; 3.99 (26 %)

Constants: Klempt et al. (2024), Table 2, converted; Fritsch et al. (2025), case I; $\alpha_0^* = 100$ from $T^* = 1/3$ (Klempt et al. 2026, case 4).

- The antibiotic lowers the living fraction of the seed from 0.965 to 0.152 (separate) and from 0.961 to 0.038 (mixed); the composition changes by less than 3 % with grid and time step
- The stress needs a small outer step: 0.025 gives about 6 % less than the limit (separate example)

The front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$ is an advection

$$\mathrm{Pe} = \frac{rh}{2\beta} \approx 31$$

on 16^3 ($r = 10\,\mathrm{mm}/T^*$, $\beta = 0.02\,\mathrm{mm}^2/T^*$). With a central gradient ϕ_{3D} oscillates and the element stops after two increments; a smaller time step does not help.

Constant added diffusion $rh/4$

First order. Stable in the element, but the stress on 16^3 is about a quarter of the converged value.

Lax–Wendroff term with an extremum sensor; upwind gradients

Second order away from extrema. In the element the Lax–Wendroff form matches Python on 16^3 but is unstable on 24^3; upwind MLS gradients are stable on 8^3 to 24^3.

TVD upwinding (MUSCL, minmod)

Used in Python for the main results; converged from 64^3 on.

Grid convergence: the stress needs a second-order front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$

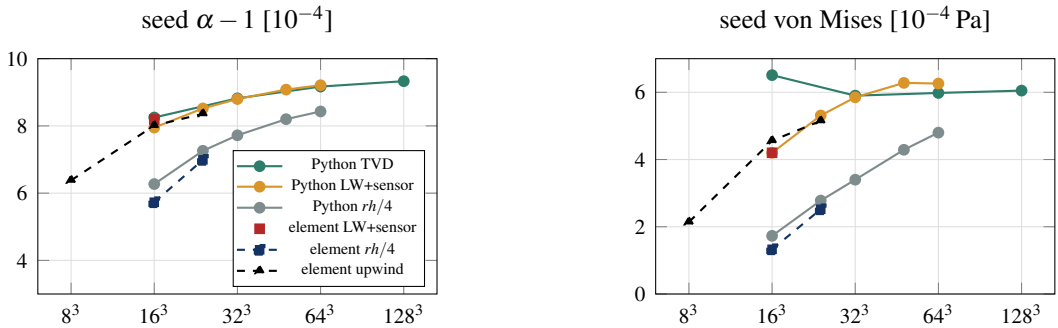


Figure 4: Seed means of the separate example with antibiotic at $T^* = 1$ against the grid ($h = 2\,\mathrm{mm}/n$). Element: your element with the agreed coupling (ANSYS). With upwind gradients its stress extrapolates to 5.8×10^{-4} Pa (GCI 16.5 %), within the band of the second-order schemes.

What the comparison found in the element and in Python

Surface points of the element

ϕ at the surface points was an inverse-distance mean that also took inner points; at a wall with $\phi = 1$ it came out low and ϕ leaked through the wall. Taking the nearest Gauss point reduced the leak to a quarter.

Neighbour selection

The search keeps the first 30 points in element order; on 16^3 and 24^3 the stencils became asymmetric inside the domain. Selecting by distance makes them symmetric to 10^{-15}.

Column sums of the NEM Laplacian

The rows sum to zero, the columns do not, so the operator does not conserve ϕ with equal weights. A conservative form removes the rest of the leak.

Python wall gradient

My Python version took a one-sided gradient in the wall cells for the front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$; corrected to the zero-flux mirror. After both corrections the codes agree on 8^3 within 1 to 5 %.

Growth around rigid bars (Klempt 2024, Sect. 4.3)

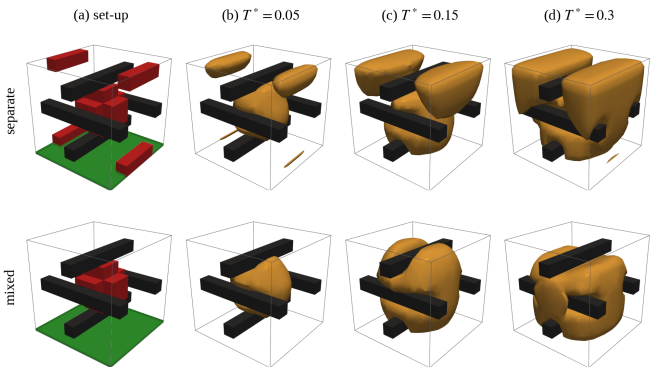


Figure 5: $\phi_{3D} \geq 0.5$ around four rigid bars, separate (top) and mixed (bottom) start; set-up after Klempt et al. (2024), Fig. 9.

- Four bars after Fig. 9 of the paper (positions read off the sketch), no biofilm or nutrient inside, frictionless contact
- The biofilm grows around the bars and closes behind them, as in Fig. 10
- The bars restrain the swelling: seed stress about ten times higher than without bars

The maze of Fig. 11 with the coupling

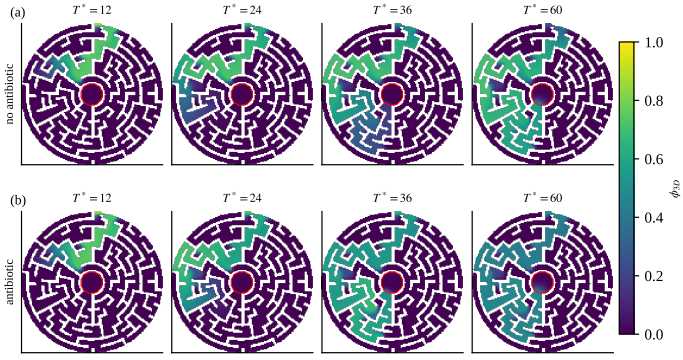


Figure 6: ϕ_{3D} in the maze of Klempt et al. (2024), Fig. 11 (560 µm across, 2D, $h = 4$ µm, Table 6 constants), without (a) and with (b) antibiotic held at the entrance from $T^* = 20$. Nutrient held in the centre (red), colony at the entrance (top).

The biofilm follows the corridors, also into dead ends, and comes within about 40 µm of the source by $T^* = 60$; with the antibiotic it stays about 130 µm away.

Which composition does new biofilm get?

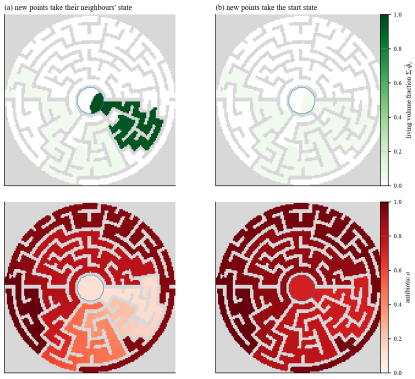


Figure 7: Maze at $T^* = 60$ with antibiotic: $\sum_i \bar{\phi}_i$ (top) and a (bottom). (a) New points take the living composition of their neighbours, (b) new points take the start state.

- With the neighbours' living composition the biofilm recovers behind the antibiotic: living volume fraction 0.35 at the end, against 0.04 when new points start from the start state
- In the cube examples the two rules differ by a few percent
- I use the neighbours' living composition as the base case: the biofilm spreads because the living cells grow

Questions and next steps

Questions for you

1. How was the front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$ stabilised in the runs of Klempt 2024? The cell Péclet number is about 25 on the paper's mesh.
2. Can I keep the three element fixes (nearest Gauss point at the surface, conservative Laplacian, neighbours by distance)?
3. Is a stabilised front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi_{3D}$ acceptable in the thesis, stated as a numerical treatment of Eq. 34?

Questions for Felix

- The factor 1/2 in Eq. 19 of Klempt et al. (2026)
- The $+\gamma$ term in Eq. 17
- The stabilisation used in Sect. 4.2 of Klempt et al. (2024)
- Which composition new biofilm should take

Limitation

- The front speed of Eq. 34 depends on the nutrient only, not on the living biomass: after the antibiotic, dead biofilm still spreads (two thirds of the newly reached cells in the maze)

Next steps

- Thesis finished by 4 November with the results up to then
- Thesis PDF by 14 December, colloquium on 16 December
- A joint paper on the coupling and its verification

References

F. Klempt, M. Soleimani, P. Wriggers, P. Junker. A Hamilton principle-based model for diffusion-driven biofilm growth. *Biomech. Model. Mechanobiol.* 23 (2024) 2091–2113.

F. Klempt, H. Geisler, M. Soleimani, P. Junker. A continuum multi-species bacterial growth model with a novel interaction scheme. *Arch. Appl. Mech.* 96 (2026) 164.

L. Fritsch, H. Geisler, J. Grashorn, F. Klempt, M. Soleimani, M. Broggi, P. Junker, M. Beer. Bayesian updating of constitutive parameters under hybrid uncertainties with a novel surrogate model applied to biofilms. arXiv:2512.15145 (2025).

Appendix A: notation

$\phi_{3D},\ \alpha$	biofilm phase field and growth ($\mathbf{F}_g = \alpha\mathbf{I}$, $\alpha(0) = 1$), Klempt et al. 2024
$c,\ a$	nutrient and antibiotic fields (first and second nutrient field of the element)
$\beta,\ k_\alpha,\ r,\ k$	diffusion of ϕ_{3D}, growth rate, front speed, Monod constant (Eqs. 34, 36)
$d,\ g$	diffusion and uptake of c and a (Eq. 35)
$\phi_i,\ \psi_i,\ \phi_0$	volume fraction and living fraction of species i, void (Klempt et al. 2026)
$\bar{\phi}_i = \phi_i\psi_i$	living volume fraction of species i; $\sum_i \bar{\phi}_i$ enters the consumption
$c^* = c_0^* c,\ \alpha^* = \alpha_0^* a$	nutrient and antibiotic seen by the point model (not the growth α)
$T^*,\ s$	normalised time; point-model time per unit T^* ($s = 0.1$)

Appendix B: equations and constants

$$\dot{\phi}_{3D} = \beta \nabla^2 \phi_{3D} + k_\alpha \alpha - r \tfrac{c}{k+c} \mathbf{n}_c \cdot \nabla \phi_{3D} \qquad \text{(Eq. 34 with the front term } -r\tfrac{c}{k+c}\mathbf{n}_c \cdot \nabla\phi_{3D}\text{)}$$

$$\dot{c} = d\nabla^2 c - g\,\phi_{3D} \textstyle\sum_i \bar{\phi}_i\, c, \qquad \dot{a} = d\nabla^2 a - g\,\phi_{3D} \textstyle\sum_i \bar{\phi}_i\, a \qquad \text{(Eq. 35; } g\phi \text{ printed is a misprint)}$$

$$\dot{\alpha} = k_\alpha\, \phi_{3D} \qquad \text{(Eq. 36, unchanged)}$$

Point model of Klempt et al. 2026 at every Gauss point as published, with $\sum_i \phi_i + \phi_0 = 1$.

β	0.02 mm$^2/T^*$	Klempt 2024 Table 2, converted to the 2 mm cube (assumption)
d, g	10^8 mm$^2/T^*$, $g\ell^2/d = 1$	Table 2 (fields quasi-steady within $4 \times 10^{-8}\, T^*$)
r, k, k_α	10 mm/T^*, 1, 10^{-3} 1/T^*	Table 2
E, ν	10 Pa, 0.49	Table 2; void floor $f = 10^{-3}$ (converged)
point model, c_0^*, s	case I, 100, 0.1	Fritsch et al. 2025
α_0^*, onset	100 from $T^* = 1/3$	Klempt et al. 2026 case 4
$\phi_{\min}$	0.001	not in a paper; converged (assumption)